

Biochemistry #1B
Extinction Coefficient of Methylene Blue
Matthew Hale, Mai Tomosaka

Introduction:

This experiment aimed to determine the molar extinction coefficient of methylene blue by constructing a curve of absorbance vs. concentration. The absorbances of the dilutions of methylene blue were measured at 663 nm. The resulting data was used to calculate an extinction coefficient and was compared to the literature value of methylene blue.

Materials and Methods:

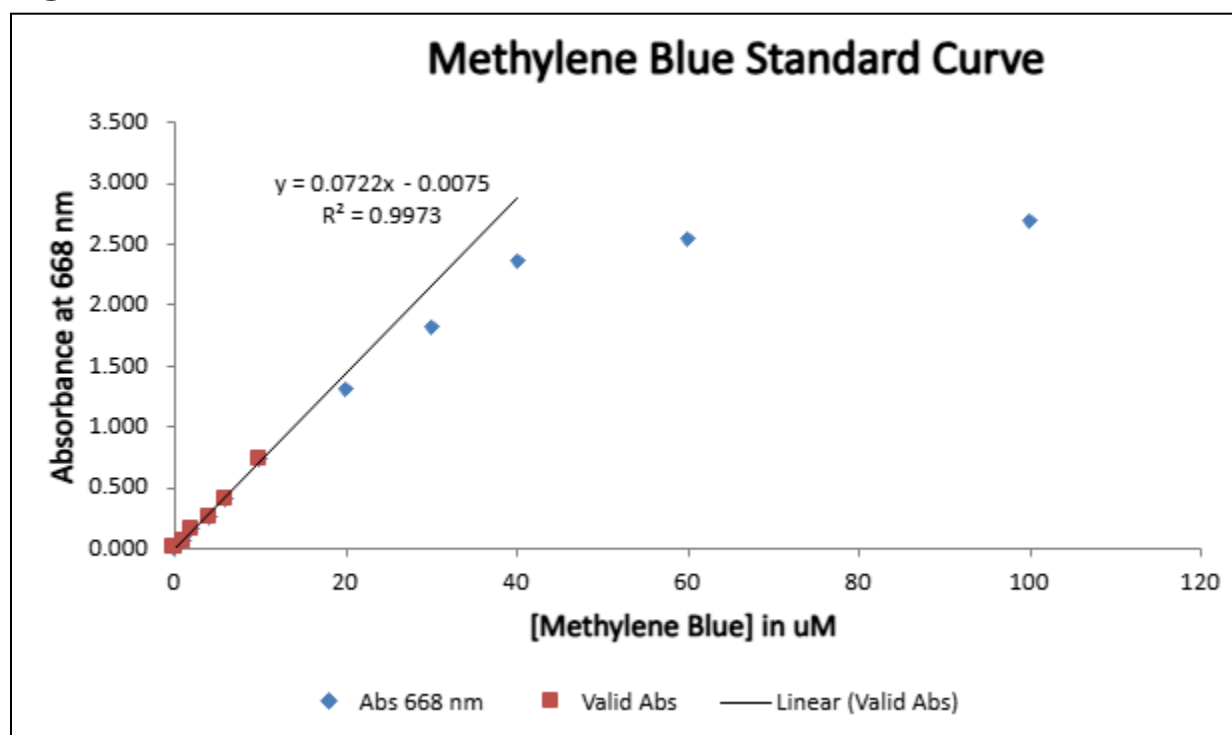
A 0.5% methylene blue stock solution was provided. Serial dilutions were prepared (Table 1) and their absorbances were measured in a 3mL cuvette using a Shimadzu MiniUV spectrophotometer.

Results and Discussion:

A 100 μM solution was prepared using the 0.5% methylene blue stock provided. Serial dilutions of the prepared solution were used to create a standard curve of the absorbance of the dye from the spectrophotometer data recorded at 663 nm (Figure 1). The standard curve for the data recorded is accurate from approximately 1-10 μM . This data, known as the valid absorbance range, can be used to calculate the molar extinction coefficient for methylene blue. Using the data from the valid absorbance on the standard curve, the extinction coefficient for methylene blue at 663 nm is $6.79 \times 10^4 \text{ L mol}^{-1} \text{ cm}^{-1}$. This is approximately 7.1% lower than the literature value reported from OMLC by Prahl at $7.30 \times 10^4 \text{ L mol}^{-1} \text{ cm}^{-1}$. The experiment should be accurate within about 10%, indicated by the linear correlation of the valid absorbance data points. The most probable source of error is the preparation of the 0.5% methylene blue stock solution, since it is only reported to one significant figure (20% relative error). The purity of the dye was also not provided. Within this given margin of error, the experimental extinction coefficient aligns with the literature value reported by Prahl.

Table 1: Volume, Concentration, and Absorbance of Methylene Blue

Volume Methylene Blue Stock (mL)	Methylene Blue Concentration (μM)	Absorbance (668 nm)	Valid Absorbance (668 nm)
0.00	0	0.000	0.000
0.01	0.2	0.011	0.011
0.05	1	0.059	0.059
0.10	2	0.151	0.151
0.20	4	0.262	0.262
0.30	6	0.410	0.410
0.50	10	0.729	0.729
1.00	20	1.303	-
1.50	30	1.809	-
2.00	40	2.362	-
3.00	60	2.544	-
5.00	100	2.687	-

Figure 1:**References:**

Results from Scott Prahl, <https://omlc.org/spectra/mb/mb-water.html>

Question 1:

What is the range of methylene blue concentration that the spectrophotometer can accurately measure? In other words, what are the smallest and largest concentrations that can be measured with an accuracy of about 10%?

Answer: The spectrophotometer will only be accurate in the linear region of Beer's law. The data indicate that this range is from 1 μM to 10 μM methylene blue. Below 1 μM , there is too much noise from the instrumentation. Above 10 μM , the relationship begins to deviate from the linear slope, as this is the instrument's capacity.

Question 2:

It is not uncommon in this lab for different groups, using exactly the same 0.5% stock solution, to get very good correlation coefficients for their data (i.e., > 0.998) and yet have differences in their molar extinction coefficients of 5-10%. What is the simple explanation for these differences?

Answer: Since the uncertainty of our concentration of the 0.5% methylene blue stock solution is only reported to one significant figure (20% relative error). The extinction coefficient depends directly on the concentration of the substance. Even small differences in the actual concentration of the stock solution would explain the differences between groups.

Question 3:

You can use error analysis in Excel to determine the 95% confidence range for your extinction coefficient. To do this, you need to load the add-in packages (under the tools tab on Macs), go ahead and load both the AnalysisTool-Pac and Solver, which we use later in the semester. On a Mac, return to the tools table and select Data Analysis, then choose regression. Input the Y and X column data for your valid data, and you will get the analysis, usually on a new Excel sheet. Look for the 95% confidence results for your X variable and calculate the difference between the high (or low) value and your slope (ie, your error) and convert to a % of your slope.

Answer: From the linear regression in Excel, the 95% confidence interval is obtained from the X variable output. The error is the difference between the slope and the upper bound of the interval, then expressed as a percentage of the slope. In our case, the 95% confidence interval for the extinction coefficient was 6.79×10^4 to $7.65 \times 10^4 \text{ Lmol}^{-1}\text{cm}^{-1}$. This aligns with the uncertainty of $\pm 6\%$ relative to the best fit value of $7.22 \times 10^4 \text{ Lmol}^{-1}\text{cm}^{-1}$. The literature value also falls within this confidence interval.